

WASP-43b

R-0484

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230426 · created 2026-07-20T07:47:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460060.7359 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.059
Transit depth (Rp/Rs)^2	3.68 +/- 2.26 [%]
Orbital Inclination (inc)	81.79 +/- 2.86
Airmass coefficient 1 (a1)	0.84 +/- 0.22
Airmass coefficient 2 (a2)	0.09 +/- 0.12
Scatter in the residuals of the lightcurve fit is	25.13 %
Best Comparison Star	#3 - [111, 388]
Optimal Aperture	1.53
Optimal Annulus	6.26
Transit Duration (day)	0.049 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.059 vs 0.1594 (deviation 0.55σ)
Duration fit vs published	0.049 d vs 0.0512 d (deviation 0.17σ)
Mid-time O–C	4.6 min
Depth SNR	3.2
Residual scatter	25.13 %

Light curve

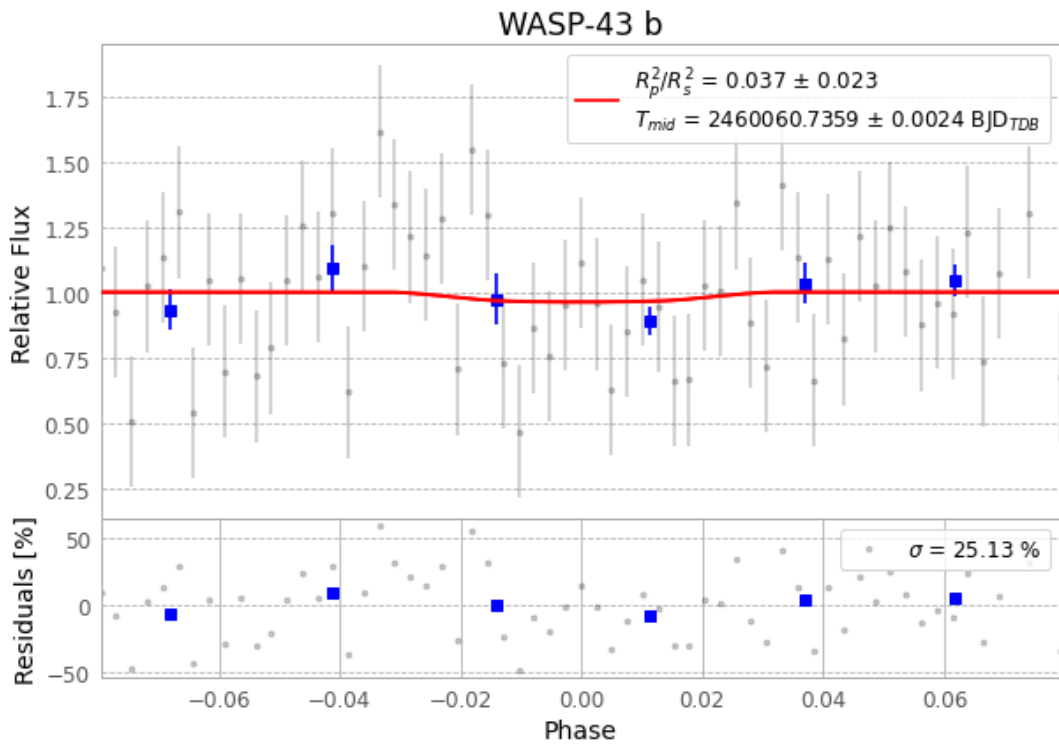


Figure 1 — FinalLightCurve_WASP-43 b_2023-04-25.png (SHA-256 9b885b207c602961...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [567, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e8c9ea73e47530ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

62 FITS frames (41 MB), WASP-43230426040015.FITS → WASP-43230426070615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230426.log (2dbb5d0c90be...), FinalLightCurve_WASP-43 b_2023-04-25.png (9b885b207c60...), FinalLightCurve_WASP-43 b_2023-04-25.csv (2091f3ff92ac...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 07:47Z.